Rahul Sangaramahalingam

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LinkedIn | GitHub | Portfolio

Summary

First-year M.S. student in Computer Science and Engineering at NSYSU, with a B.Tech in CSE and four peer-reviewed papers in cryptography, authentication, and digital forensics. I like building practical security tools, from an IoT vulnerability scanner to phishing detection and file forensics. I am looking for thesis research in network security and attack detection, and I would be glad to talk about where I could contribute.

Experience

Python Intern, Appin Technologies, Coimbatore, India — Jun to Jul 2024

- Worked on Python-based security tooling and scripting under supervision

Technical Skills

Core Areas: Network and IoT security, digital forensics, cryptographic protocols, authentication systems
Security Tools: Wireshark, Nmap, Metasploit, Burp Suite, Kali Linux, Autopsy, FTK Imager
Languages: Python, JavaScript, TypeScript, Java, PHP
AI: LLM Integration (Gemini API, Groq AI)
Platforms: Linux, AWS, Docker, Git, Supabase
Web & Backend: React, React Native, Node.js, Express.js, REST APIs, MySQL

Projects

IoT Device Security Analyzer (IDSA) — Hackhustle'25: Top 10, Internship Offered
Python, Tkinter, CVE/NVD, CVSS

- USB and network discovery via ping sweeps and port scans; checks default credentials and exposed services
- Real-time CVE/NVD integration with CVSS-based risk scoring for IoT and host vulnerability scanning
- Generates HTML/PDF/JSON reports mapped to NIST and ISO 27001 frameworks
- Tkinter GUI designed for accessibility to technical and non-technical users

SecuBrowse: Modular Browser Extension for Safe Browsing — Springer, 2026
JavaScript, Manifest V3, Gemini API

- Manifest V3 Chrome extension using the Gemini API for real-time phishing and malicious site detection
- Detects phishing, malicious sites, and social engineering via domain and script behavior analysis

Optimized Framework for Deleted File Recovery — Unpublished, 2025
Python, Random Forest, FAT32, NTFS, EXT4

- Signature-based file carving combined with a Random Forest classifier across FAT32, NTFS, EXT4
- Classifier trained to distinguish recoverable fragments from noise, improving on pure signature-matching
- Deployed as a standalone Windows application for practical forensic use

Quantum-Resistant IAM System — IEEE, 2025
Python, REST APIs, RBAC, MFA, post-quantum cryptography

- Enterprise identity and access management system integrating RBAC, MFA, and post-quantum cryptographic primitives
- Exposed via REST APIs for integration into existing enterprise systems
- Evaluated integration of quantum-resistant mechanisms while maintaining compatibility and scalability

Secure File Management System
PHP, MySQL

- Web-based platform in PHP and MySQL with RBAC, encrypted file storage, and user authentication
- Implemented secure upload/download, access authorization, activity monitoring, and audit logging

HireSight Mobile 2026

React Native, TypeScript, Groq AI, Supabase, AWS

- AI-integrated mobile application built on Supabase and AWS backend infrastructure
- Uses Groq AI for intelligent processing within a cross-platform React Native app

CineCurator Mobile 2026

React Native, Expo, Trakt API, YouTube API

- Cross-platform mobile app integrating Trakt and YouTube APIs for curated content discovery

Education

National Sun Yat-sen University (NSYSU), Kaohsiung, Taiwan Sep 2026 to present

M.S., Institute of Computer Science and Engineering | Fall 2026 courses: Cryptography

Kalasalingam Academy of Research and Education, India 2022 to 2026

B.Tech, Computer Science and Engineering CGPA: 8.29/10

Coursework: Network Security, Cryptography, Machine Learning, Operating Systems, Cloud Computing, Data Structures & Algorithms

National Chung Cheng University (CCU), Taiwan Sep 2024 to Jan 2025

Semester Exchange Program Overall: 87%

Courses: Computer Networks, System & Software Security, Operating Systems, Investment Analysis, Chinese for Beginners (93/100), Psychology: Neuroscience

Publications & Certifications

1. **SecuBrowse:** Browser extension on Manifest V3 using the Gemini API for real-time phishing and malicious site detection via content and behavioral analysis. *Springer, 2026.*
2. **Anti-Remote Disk Imaging:** Host-based defense detecting and blocking unauthorized remote disk-imaging; achieved 92% prevention rate. *IEEE, 2024.*
3. **Website Authenticator:** Phishing-resistant TOTP-based authentication scheme to reduce credential theft from spoofed login pages. *IEEE, 2024.*
4. **Quantum-Resistant IAM:** IAM architecture combining RBAC, MFA, and post-quantum cryptographic primitives, exposed via REST APIs. *IEEE, 2025.*

Certifications: Cyber Security Fundamentals (Teachnook) | Cybersecurity & Privacy (NPTEL) | Software Testing (NPTEL) | Soft Skill Development (NPTEL)

Achievements

Hackhustle'25 - Team Lead, Top 10 Finalist (IoT Device Security Analyzer, led to an internship offer) 2025

Techfrontier'25 - Team Lead, Participant 2025

Languages

Tamil	Native	**English**	Professional working proficiency
Hindi	Conversational	**Mandarin Chinese**	Beginner

Mandarin: 93/100 in Chinese for Beginners at CCU | 730+ day Duolingo streak in Mandarin as of Nov 2026

References

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